AI Powered Fake Image and Video Detection System

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Agenda

Introduction

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- Deepfakes are AI-generated manipulated images and videos.
- They pose risks in misinformation, fraud and identity theft.
- Our project develops a CNN-based detection system with future video support.

Motivation

- Increasing misuse of generative AI
- Fake news propagation
- Financial fraud
- Social media manipulation
- Need for automated detection

Problem Statement

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Develop an AI system capable of distinguishing real and fake images accurately and extend the framework for fake video detection.

Objectives

1. Detect fake images using CNN.
2. Achieve high classification accuracy.
3. Deploy as a web application.
4. Extend to video deepfake detection.
5. Improve robustness using advanced models.

Literature Review

Literature Review

1. **Lightweight AI-Generated Image Detection Based on Frequency Features**
   Shuo Li et al.
   Uses frequency-domain features for efficient AI-generated image detection with low computational cost.

2. **DeepFake Detection in the AIGC Era: A Survey**
   Shuaijie Xie
   Reviews deepfake detection techniques for GAN- and diffusion-generated images, challenges, and future directions.

3. **Explainable Deepfake Detection: A Multi-Model Framework for Image Forensics**
N. Bharati
Integrates Explainable AI (XAI) with deep learning to improve interpretability of deepfake detection.

4. **Methods and Trends in Detecting Generated Images: A Comprehensive Review**
Arpan Mahara, Naphtali Rishe
Reviews datasets, detection methods, evaluation metrics, and future research trends.

Literature Review Comparison

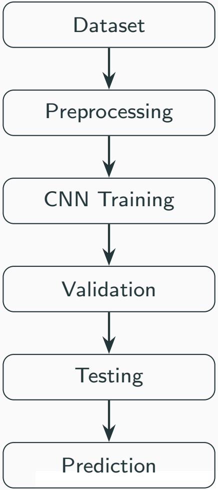
Paper	Strength	Weakness	Application
Frequency Features (Shuo Li et al.)	Lightweight and computationally efficient	Less effective against advanced generative models	Real-time AI image detection
AIGC DeepFake Survey (Shuaijie Xie)	Comprehensive review of modern detection techniques	No novel detection model proposed	Research reference and benchmarking
Explainable Deepfake Detection (N. Bharati)	High interpretability using XAI	Additional computational overhead	Trustworthy image forensics
Generated Image Detection Review (Arpan Mahara et al.)	Covers datasets, metrics, and research trends	Review only; lacks implementation	Future research guidance

Literature Review

Method	Strength	Weakness	Use
CNN	Fast	Dataset dependent	Baseline
ResNet	Accurate	Heavy	Image analysis
EfficientNet	Efficient	Complex tuning	Classification
Vision Transformer	Powerful	Expensive	Large datasets
GAN Fingerprints	Robust	Generator specific	Deepfake detection

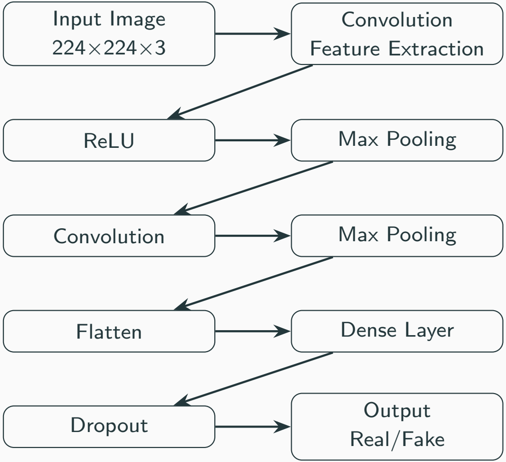
Methodology

Proposed Workflow



- **Dataset:** Collect real and fake images from publicly available datasets.
- **Preprocessing:** Resize images to 224×224, normalize pixel values, and split into training, validation, and testing sets.
- **CNN Training:** Train the convolutional neural network to learn discriminative visual features.
- **Validation:** Evaluate performance after each epoch to monitor learning and reduce overfitting.
- **Testing:** Measure the final model performance using unseen images.
- **Prediction:** Classify an uploaded image as **Real** or **Fake** through the trained model.

CNN Architecture



- **Input:** Receives a resized RGB image.
- **Convolution:** Extracts edges, textures and facial patterns using learnable filters.
- **ReLU:** Adds non-linearity, enabling the network to learn complex features.
- **Max Pooling:** Reduces feature-map size while preserving important information.
- **Flatten:** Converts feature maps into a 1-D vector.
- **Dense Layer:** Learns high-level relationships
- **Dropout:** Prevents overfitting by randomly disabling neurons during training.
- **Output:** Uses a sigmoid classifier to predict whether the image is **Real** or **Fake**.

Experimental Setup

Dataset

- **Dataset: Deepfake and Real Images**
- Image size: 224×224
- Data split: Train (8730) / Validation (2181) / Test (2181)
- Fake Images:6581
- Real Images:6511
- Normalization and augmentation

Dataset Samples

Fake Images

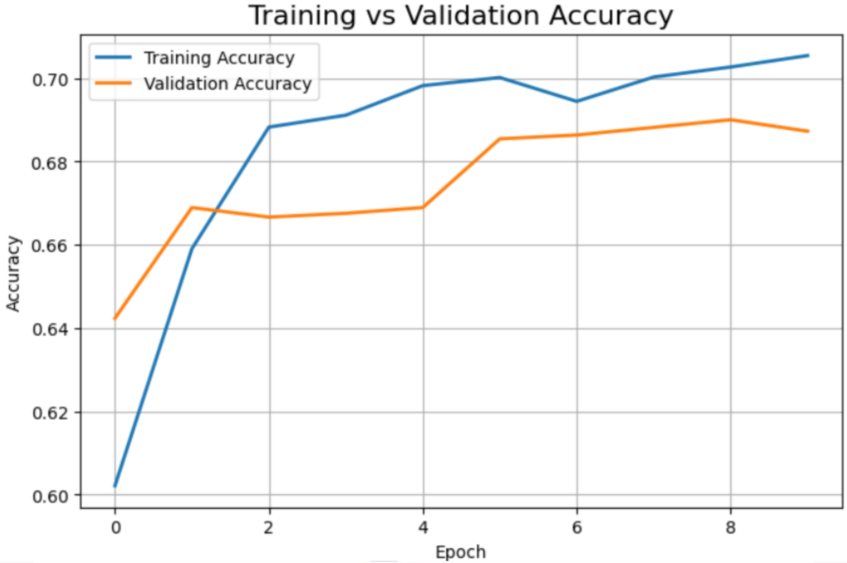
Real Images

Experimental Setup

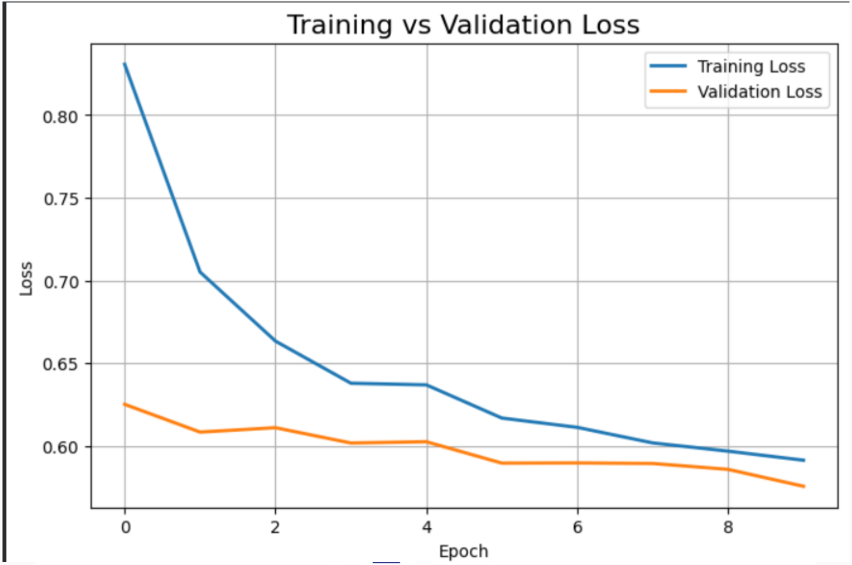
Framework	TensorFlow/Keras
Language	Python
Optimizer	Adam
Loss	Binary Cross Entropy
Epochs	10
Batch Size	32

Results

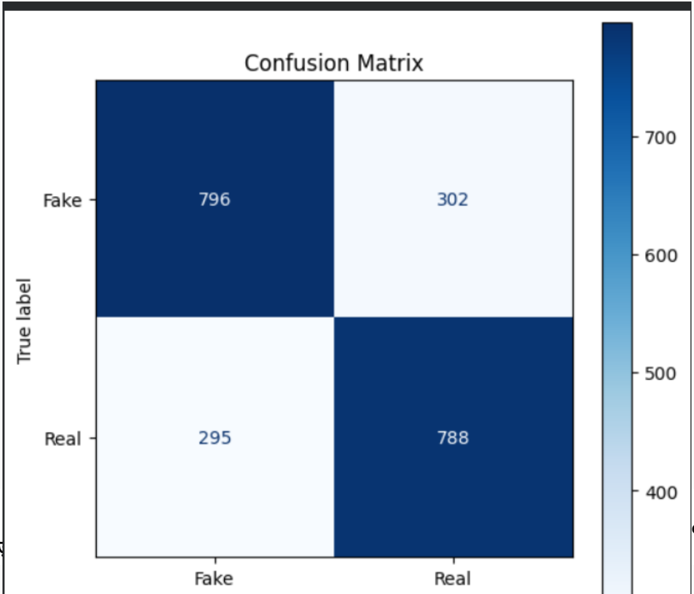
Training Accuracy



Training Loss



Confusion Matrix



Performance Metrics

```
==================================================
      MODEL EVALUATION METRICS
==================================================
Accuracy : 72.44%
Precision: 71.95%
Recall   : 72.95%
F1-Score : 72.44%
==================================================

Classification Report:

              precision    recall  f1-score   support

        Fake     0.7295    0.7195    0.7244      1098
        Real     0.7195    0.7295    0.7244      1083

    accuracy                         0.7244      2181
   macro avg     0.7245    0.7245    0.7244      2181
weighted avg     0.7245    0.7244    0.7244      2181
```

Evaluation Summary

Metric	Value
Accuracy	72.44%
Precision	71.95%
Recall	72.95%
F1-Score	72.44%

- **Accuracy:** Percentage of correctly classified images.
- **Precision:** Fraction of images predicted as fake that are actually fake.
- **Recall:** Ability of the model to correctly identify fake images.
- **F1-Score:** Harmonic mean of precision and recall, indicating overall classification performance.

Discussion

Challenges

- Dataset imbalance
- Generalization to unseen deepfakes
- Computational requirements
- Overfitting mitigation
- Image classifier to video classifier

Future Work

- Decrease video detecting latency
- Extend support for audio and GAN-generated image detection
- Explainable AI
- Real-time browser extension
- Mobile application

Conclusion

- Successfully developed a CNN-based fake image and video detector.
- Achieved 72.44% accuracy on the evaluation dataset.
- Provides a strong foundation for fake image and video detection.

References i

- FaceForensics++: Learning to Detect Manipulated Facial Images.
- He et al., Deep Residual Learning for Image Recognition.
- EfficientNet: Rethinking Model Scaling.
- GAN Fingerprints for Fake Image Detection.
- TensorFlow Documentation.

Thank You